Carbonly — Business Requirements Document

v2.0 · April 2026

Business Requirements Document (BRD)

Carbonly — Carbon Emissions Management SaaS

Version: 2.0 **Date:** April 2026 **Status:** Approved — Active Development **Author:** Carbonly Product Team

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1. Executive Summary

Carbonly is a multi-tenant SaaS platform that enables organisations to measure, track, and report greenhouse gas (GHG) emissions across **Scope 1, Scope 2, and Scope 3** as defined by the GHG Protocol Corporate Accounting and Reporting Standard. The platform provides a structured emissions calculator, an operational dashboard, automated PDF/CSV reporting, and a full platform administration layer with subscription-based access control.

Built on Next.js 16 (App Router), TypeScript, Material UI, PostgreSQL via Prisma, and Stripe for payments, Carbonly is architected as a multi-tenant shared-schema platform serving unlimited organisations ("entities") from a single deployment. Tenant data isolation is enforced at the database query layer using `entityId` discrimination, not separate schemas.

This document captures both the high-level business requirements and the detailed user stories (with validations, business rules, acceptance criteria, and edge cases) that govern every feature of the platform. It is the single authoritative reference for engineering, QA, and product stakeholders.

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2. Business Objectives

#	OBJECTIVE	SUCCESS METRIC
BO-1	Enable organisations to comply with GHG reporting requirements (TCFD, CDP, EU CSRD)	≥ 1 published annual report per active entity/year
BO-2	Reduce the time to compile annual emissions inventories from weeks to hours	Report generation completes in < 5 seconds; total inventory entry < 4 hours
BO-3	Generate recurring SaaS revenue through tiered subscription plans (Starter, Professional, Enterprise)	MRR growth ≥ 10% MoM in first 12 months
BO-4	Support multiple organisations on a single platform with strict data isolation	Zero cross-tenant data leakage incidents
BO-5	Provide configurable emission factors aligned to DEFRA, EPA, GHG Protocol, and IPCC sources	Factor library covers ≥ 95% of standard GHG Protocol activity categories

3. Scope

3.1 In Scope

- **Scope 1 emissions calculator** — stationary combustion (natural gas, diesel, heating oil), mobile combustion (petrol, diesel vehicles), fugitive emissions (refrigerants)
- **Scope 2 emissions calculator** — purchased electricity, heat/steam (location-based method)
- **Scope 3 emissions calculator** — business travel, employee commuting, purchased goods, waste
- **Emissions dashboard** — scope breakdown, monthly trends, year-over-year comparison, baseline comparison
- **Emission records management** — full CRUD, manual entry, paginated list with filtering
- **Admin panel** — entity management, user management, scope/activity configuration, emission factor management, yearly configuration
- **Subscription and payment management** via Stripe (Checkout, Customer Portal, Webhooks)

- **Role-based access control**: SUPER_ADMIN, ADMIN (entity-level), EXPERT (entity-level)
- **PDF/CSV report generation** per year per scope
- **Multi-entity (multi-tenant) architecture** with complete data isolation
- **Marketing landing page** with animated sections, pricing, and onboarding CTAs

3.2 Out of Scope — Phase 1

- Automated data ingestion via IoT sensors or utility APIs
- Third-party ESG platform integrations (Salesforce, SAP, Microsoft Sustainability Manager)
- Market-based Scope 2 accounting (RECs, PPAs, guarantees of origin)
- Carbon offsetting marketplace
- Native mobile application (iOS / Android)
- Entity-specific emission factor overrides via yearly config UI (data model supports it; UI deferred)
- Bulk CSV import of emission records

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4. Stakeholders

STAKEHOLDER	PLATFORM ROLE	SYSTEM ROLE
Platform Owner	Carbonly product team	Defines features, roadmap, pricing
SUPER_ADMIN (Adam)	Platform administrator	Configures taxonomy, monitors tenants, views billing analytics
Entity ADMIN (Emma)	Customer organisation admin	Manages users, yearly config, billing, publishes reports
Carbon Expert / EXPERT (Carlos)	End user at customer org	Enters activity data, runs calculations, views dashboard
Visitor / Prospect (Priya)	Unauthenticated visitor	Discovers platform, evaluates pricing, registers
Finance Team	Customer organisation	Approves subscription, manages payment methods

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5. Functional Requirements

5.1 Authentication & Authorisation

ID	REQUIREMENT	PRIORITY
FR-AUTH-01	Users must authenticate via email/password or Google OAuth	Must
FR-AUTH-02	Sessions must expire after inactivity (configurable, default 30 days JWT)	Must
FR-AUTH-03	SUPER_ADMIN has access to the entire platform admin panel (`/admin/**`)	Must
FR-AUTH-04	Entity ADMIN can manage users and configuration within their entity	Must
FR-AUTH-05	EXPERT can only access and create emission records within their assigned entity	Must
FR-AUTH-06	Users not assigned to an entity must be redirected to an onboarding flow	Must
FR-AUTH-07	All protected routes must be guarded by middleware at the edge (no client-only guards)	Must
FR-AUTH-08	Password storage must use bcryptjs (cost factor 10); plain text never persisted	Must

5.2 Emissions Calculator

ID	REQUIREMENT	PRIORITY
FR-CALC-01	The system must provide calculators for Scope 1, 2, and 3	Must
FR-CALC-02	Each calculator must allow selection of activity and emission factor from the taxonomy	Must
FR-CALC-03	Calculation formula: `CO2e (kg) = quantity × emission factor value`	Must
FR-CALC-04	Results must be stored in both kg CO_2e and tCO_2e	Must
FR-CALC-05	Each record must capture: activity, emission factor, quantity, unit, period (month/year), notes	Must
FR-CALC-06	Emission factors must display source, region, GWP version, and value to users	Should
FR-CALC-07	The selected emission factor ID is immutable once a record is saved	Must
FR-CALC-08	Period cannot be set in the future	Must
FR-CALC-09	Calculations are performed server-side; client input is never trusted	Must

5.3 Dashboard

ID	REQUIREMENT	PRIORITY
FR-DASH-01	Display total tCO_2e by scope for the selected year in a hero card	Must
FR-DASH-02	Show scope breakdown as an interactive pie/donut chart	Should
FR-DASH-03	Show monthly emission trend as a bar chart (12-month view)	Should
FR-DASH-04	List top 5 emission-contributing activities	Should
FR-DASH-05	Allow year selection (current year default; 2021–2026)	Must
FR-DASH-06	Show year-over-year (YoY) and vs-baseline percentage comparisons	Should
FR-DASH-07	Scope cards must be clickable to reveal a drill-down of activities	Should
FR-DASH-08	Display total consumption breakdown grouped by unit	Could
FR-DASH-09	All dashboard data is rendered server-side (RSC) on initial load	Must

5.4 Admin — Entity Management

ID	REQUIREMENT	PRIORITY
FR-ENT-01	SUPER_ADMIN can create, view, update, and deactivate entities	Must
FR-ENT-02	Each entity has a unique slug used for tenant routing	Must
FR-ENT-03	Entity ADMIN can update entity profile (name, industry, country, logo)	Should
FR-ENT-04	Subscription status must reflect Stripe state in real time via webhooks	Must
FR-ENT-05	Entity deletion must cascade to all child records (users, records, reports)	Should

5.5 Admin — User Management

ID	REQUIREMENT	PRIORITY
FR-USR-01	SUPER_ADMIN can view all users across all entities	Must
FR-USR-02	SUPER_ADMIN can create users and assign them to an entity	Must
FR-USR-03	SUPER_ADMIN can change a user's global role and entity role	Must
FR-USR-04	A user cannot delete their own account	Must

5.6 Admin — Scope & Activity Configuration

ID	REQUIREMENT	PRIORITY
FR-SCOPE-01	SUPER_ADMIN can create, update, and deactivate activities under each scope	Must
FR-SCOPE-02	Each activity must have a name, unit, and equation description	Must
FR-SCOPE-03	Deleting an activity that has emission factors must be blocked (409)	Must
FR-SCOPE-04	Deactivated activities must not appear in calculator dropdowns but must not break history	Must

5.7 Admin — Emission Factor Management

ID	REQUIREMENT	PRIORITY
FR-EF-01	SUPER_ADMIN can add emission factors to activities with name, value, unit, source, region, GWP	Must
FR-EF-02	Only one factor per activity may be marked `isDefault = true` at a time	Must
FR-EF-03	Setting a new factor as default auto-unsets the previous default for the same activity	Must
FR-EF-04	Deleting a factor referenced by any emission record must be blocked (409)	Must

5.8 Yearly Configuration

ID	REQUIREMENT	PRIORITY
FR-YEAR-01	Each entity can have a yearly configuration per calendar year	Could
FR-YEAR-02	One year can be designated as the baseline year	Could
FR-YEAR-03	Yearly config anchors entity-specific emission factor overrides (Phase 2)	Won't (Phase 1)

5.9 Payment Module

ID	REQUIREMENT	PRIORITY
FR-PAY-01	Users can subscribe to Starter, Professional, or Enterprise plans	Must
FR-PAY-02	Subscription is initiated via Stripe Checkout with a 14-day free trial	Must
FR-PAY-03	Webhook at `/api/webhooks/stripe` must update entity subscription status on all payment events	Must
FR-PAY-04	Entities with PAST_DUE or CANCELED status must surface a banner with a billing upgrade link	Must
FR-PAY-05	Entity ADMIN can view billing history and manage payment methods via Stripe Customer Portal	Must
FR-PAY-06	Stripe webhook signature must be verified before any DB mutation	Must

5.10 Reporting

ID	REQUIREMENT	PRIORITY
FR-RPT-01	Users can generate a yearly emissions report per entity	Must
FR-RPT-02	Reports must include: total emissions, scope breakdown, monthly trend, top activities, YoY, vs-baseline	Must
FR-RPT-03	Reports can be exported as CSV (PDF export: Phase 2)	Must
FR-RPT-04	Generated reports are stored with a status lifecycle: DRAFT → GENERATED → PUBLISHED	Must
FR-RPT-05	If a report for the same entity/year already exists, a 409 must be returned	Must
FR-RPT-06	Attempting to generate a report for a year with zero records returns 422	Must

6. Non-Functional Requirements

ID	CATEGORY	REQUIREMENT
NFR-01	Security	All routes protected by session-based auth; tenant `entityId` isolation enforced at query level
NFR-02	Performance	Dashboard renders initial data server-side (RSC); no client waterfalls on first load
NFR-03	Performance	Dashboard data query < 50ms on ~2,000 records; initial page load < 1.5s on fast 3G
NFR-04	Availability	Platform target 99.9% uptime SLA for Enterprise plan
NFR-05	Scalability	Multi-tenant shared schema; must support 500+ entities without schema changes
NFR-06	Auditability	All emission records capture `userId`, `dataSource`, and `createdAt`
NFR-07	Compliance	Emission factor methodology aligned to GHG Protocol; source and vintage (year) traceable
NFR-08	Accessibility	All interactive elements keyboard-accessible; colour not sole indicator of state
NFR-09	Responsiveness	All pages render correctly at 375px (mobile), 768px (tablet), 1280px (desktop)
NFR-10	Data integrity	Emission factor references in records are immutable post-creation
NFR-11	Error handling	All API routes return `{ error: string }` JSON with appropriate 4xx/5xx status codes

7. Subscription Plans

FEATURE	STARTER	PROFESSIONAL	ENTERPRISE
Monthly price	£49/mo	£149/mo	Custom
Annual price	£39/mo (billed annually)	£119/mo (billed annually)	Custom
Users	Up to 3	Up to 15	Unlimited
Scopes	1 & 2	All (1, 2, 3)	All
Emission history	1 year	Unlimited	Unlimited
Reports	Basic CSV	PDF + CSV	Custom
Support	Email	Priority email	Dedicated CSM
SLA	—	—	99.9%
Free trial	14 days	14 days	—

Billing note: Plans are managed via Stripe price IDs configured in environment variables. The "Most Popular" badge is applied to the Professional plan.

8. Assumptions & Constraints

- Phase 1 uses **location-based** Scope 2 methodology only (market-based RECs, PPAs deferred)
- Emission factors are maintained manually by the platform SUPER_ADMIN (auto-sync from DEFRA/EPA APIs deferred)
- All monetary values are in **GBP** (configurable for future markets)
- The platform is deployed on **Vercel** with serverless PostgreSQL (Neon / Supabase compatible)
- `NEXTAUTH_SECRET` must be set; Google OAuth credentials are optional (credentials-only auth still works)
- Stripe is in **test mode** for development; production mode requires live API keys
- The baseline year is **2021** by convention (configurable per entity in Phase 2)
- Maximum quantity precision is **15 significant digits** (JavaScript number safety)

- bcryptjs hashes only the **first 72 bytes** of a password (known limitation; documented)

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9. User Stories & Acceptance Criteria

This section captures every user-facing behaviour on the Carbonly platform. Each story follows the **As a / I want / So that** format, sized to fit in a single sprint, and includes explicit validation rules, business rules, Given/When/Then acceptance criteria, and edge cases.

Story conventions

FIELD	DESCRIPTION
ID	Unique identifier (e.g. `US-001`)
Priority	Must / Should / Could / Won't (MoSCoW)
Story points	Fibonacci estimate (1, 2, 3, 5, 8, 13)
Validations	Input rules enforced by UI, API, and/or database
Business rules	Non-validation domain rules
Acceptance criteria	Given / When / Then scenarios that must pass
Edge cases	Unusual paths that must be handled correctly

9.1 Personas

Adam — Platform Super Admin

- **Role:** `SUPER_ADMIN`
- **Context:** Works at Carbonly, owns the platform taxonomy and tenant lifecycle
- **Goals:** Onboard customers, maintain emission factors, monitor platform revenue
- **Routes used:** `/admin/**`

Emma — Entity Administrator

- **Role:** `ADMIN` within an Entity
- **Context:** Head of Sustainability at Acme Corp
- **Goals:** Manage team, configure yearly baseline, approve billing, publish annual reports

- **Routes used:** `/dashboard`, `/reports`, `/settings`

Carlos — Carbon Expert

- **Role:** `EXPERT`
- **Context:** Environmental consultant embedded at Acme Corp
- **Goals:** Enter monthly activity data, run calculations, verify numbers before reporting
- **Routes used:** `/scope/1`, `/scope/2`, `/scope/3`, `/emissions`, `/dashboard`

Priya — Visitor / Prospect

- **Role:** Unauthenticated
- **Context:** Discovers Carbonly via ad or referral
- **Goals:** Understand the product and decide whether to sign up
- **Routes used:** `/` (landing page), `/register`

9.2 Epic 1 — Authentication & Account Management

Goal: Allow users to securely access the platform and manage their credentials.

US-001 · Register a new account

As a visitor **I want** to create a Carbonly account **So that** I can sign in and use the platform

- **Persona:** Priya · **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Full name	Required · 2–80 chars · letters, spaces, hyphens, apostrophes
Email	Required · valid RFC 5322 format · unique across `User.email`
Password	Required · min 8 chars · must contain at least one letter and one digit
Confirm password	Must equal `password` field

Business rules

- Password stored as bcryptjs hash (cost factor 10). Plain text never persisted.

- Email comparison is case-insensitive (lowercased on both write and read).
- On successful registration, no session is created — user is redirected to `/login`.
- Duplicate-email attempts return `409 Conflict` with `"Email already registered"`.

Acceptance criteria

1. **Given** I am on `/register` **When** I submit valid name, email, and password **Then** a `User` row is created and I'm redirected to `/login?registered=1`.
2. **Given** `existing@example.com` already exists **When** I submit that email **Then** I see "Email already registered" and no new user is created.
3. **Given** password is `abc` **When** I submit **Then** I see "Password must be at least 8 characters".
4. **Given** JavaScript is disabled **When** I submit **Then** server-side validation still runs and errors surface.

Edge cases

- Whitespace-only name is rejected.
- Trailing whitespace on email is trimmed before saving.
- Unicode names (e.g. `Ayşe Yılmaz`) are accepted.

US-002 · Sign in with email and password

As a registered user **I want** to sign in with email and password **So that** I can access my organisation's data

- **Persona:** Carlos · **Priority:** Must · **Story points:** 3 · **Dependencies:** US-001

Validations

FIELD	RULE
Email	Required · valid format
Password	Required · 1+ chars

Business rules

- Authentication handled by NextAuth v5 CredentialsProvider.
- Password comparison uses `bcrypt.compare()`.
- Failed attempts return generic error `"Invalid email or password"` (no account enumeration).
- Successful sign-in issues a JWT session cookie (HttpOnly, SameSite=Lax, signed with `NEXTAUTH_SECRET`).

- JWT payload: `{ sub, globalRole, entityId, entitySlug, entityRole }`.

Acceptance criteria

1. **Given** valid credentials **When** I submit **Then** session cookie is set and I'm redirected to `/dashboard`.
2. **Given** wrong password **When** I submit **Then** I see "Invalid email or password" and no cookie is set.
3. **Given** non-existent email **When** I submit **Then** same generic error (no enumeration).
4. **Given** I'm already signed in **When** I visit `/login` **Then** I'm redirected to `/dashboard`.

US-003 · Sign in with Google

As a registered user **I want** to sign in with Google **So that** I don't need another password

- **Persona:** Emma · **Priority:** Should · **Story points:** 2

Business rules

- Google OAuth via NextAuth `GoogleProvider` (Authorization Code flow).
- On first sign-in, `User` + `Account` rows created by PrismaAdapter.
- Same email as a credentials account → new `Account` row links Google to existing user.

Acceptance criteria

1. **Given** I complete Google consent **Then** `User` created/matched and redirected to `/dashboard`.
2. **Given** existing credentials account with same email **When** Google sign-in **Then** accounts are linked — no duplicate `User`.

US-004 · Sign out

As a signed-in user **I want** to sign out **So that** my session ends on shared devices

- **Priority:** Must · **Story points:** 1

Acceptance criteria

1. **Given** I'm signed in **When** I click "Sign out" **Then** session cookie cleared, redirected to `/login`.

2. **Given** I've signed out **When** I use the browser back button to `/dashboard` **Then** middleware redirects to `/login`.

US-005 · Protected routes enforce session

As the system **I want** to block unauthenticated access **So that** data isolation is guaranteed

- **Priority:** Must · **Story points:** 2

Business rules

- Middleware (`src/middleware.ts`) runs at the edge on every request except `/api/auth/**`, `/api/webhooks/**`, `/login`, `/register`, and public assets.
- Unauthenticated requests redirect to `/login`.

Acceptance criteria

1. **Given** no session cookie **When** I request `GET /dashboard` **Then** 307 redirect to `/login`.
2. **Given** valid session but `globalRole !== SUPER_ADMIN` **When** I request `/admin` **Then** redirect to `/dashboard`.
3. **Given** any authenticated request **When** server queries tenant data **Then** query filters by `entityId` from `session.user.entityId`.

9.3 Epic 2 — Onboarding & Workspace Navigation

Goal: Help users find their way around the app after signing in.

US-010 · Land on the dashboard after sign-in

As Carlos **I want** to see my dashboard as my landing page **So that** I get value within seconds

- **Priority:** Must · **Story points:** 1

Acceptance criteria

1. **Given** I sign in successfully **Then** I see the dashboard for the current year with total emissions and scope cards.

US-011 · Navigate via the persistent sidebar

As any user **I want** a sidebar with links to all major areas **So that** I can move between sections freely

- **Priority:** Must · **Story points:** 2

Business rules

- Permanent on desktop (≥ 992px), temporary drawer (hamburger) on mobile.
- Active route is visually highlighted.
- Sidebar items: Dashboard, All Records, Scope 1/2/3 calculators, Reports, Settings, Admin Panel (if permitted).
- Carbonly logo links to `/`.

Acceptance criteria

1. **Given** I'm on the dashboard **When** I click "All Records" **Then** I navigate to `/emissions` with the sidebar highlighting that item.
2. **Given** mobile viewport (< 992px) **When** I tap the hamburger **Then** temporary drawer slides in; tapping a link closes the drawer.

US-012 · Switch role (demo mode)

As a demo visitor **I want** to switch between admin and expert views without re-login **So that** I can explore both perspectives

- **Priority:** Could (demo-only) · **Story points:** 2

Acceptance criteria

1. **Given** signed in as SUPER_ADMIN on `/admin` **When** I click "Expert View" **Then** session replaced by the expert account and redirected to `/dashboard`.

9.4 Epic 3 — Emissions Dashboard

Goal: Give tenant users instant insight into their carbon footprint.

US-020 · View total emissions hero panel

As Emma **I want** a large "Total Emissions" card **So that** I see the headline number at a glance

- **Priority:** Must · **Story points:** 3

Business rules

- Hero shows total tCO_2e for selected year (1 decimal place).
- 12-month sparkline shows monthly distribution.
- Two comparison chips: prior year (YoY %) and baseline year (vs-baseline %).
- Chip colour: green if lower, red if higher.
- No data state: "No data yet — start by logging emissions in the Scope calculators".

Acceptance criteria

1. **Given** records totalling 1,248.5 tCO_2e for 2025 **When** I view `/dashboard?year=2025` **Then** hero shows "1,248.5 t CO_2e".
2. **Given** prior year was 1,420 tCO_2e **When** current year is 1,248.5 **Then** YoY chip reads "↓ 12%" in green.
3. **Given** 2025 is the baseline year **Then** no "vs baseline" chip is shown.

US-021 · View scope breakdown cards

As Carlos **I want** individual cards for Scope 1, 2, and 3 **So that** I see where emissions concentrate

- **Priority:** Must · **Story points:** 2

Business rules

- Three cards (Bootstrap `col-12 col-md-4`): label, sub-label, tCO_2e, % of total, progress bar.
- Cards are clickable to expand an activity drill-down below.
- Colour coding: Scope 1 green, Scope 2 teal/blue, Scope 3 amber.

Acceptance criteria

1. **Given** Scope 2 = 518.4 t of 1,248.5 total **When** card renders **Then** shows "42%" with bar at 42%.
2. **Given** I click Scope 2 **Then** panel shows Scope 2 activities sorted by CO_2e descending.
3. **Given** I click the same card again **Then** panel collapses.

US-022 · View scope breakdown pie chart

As Emma **I want** an interactive pie chart **So that** I have a visual proportion at a glance

- **Priority:** Should · **Story points:** 3

Business rules

- Recharts `PieChart` with 3 slices. Hover shows tooltip with tCO_2e. Click slice → drill-down.

US-023 · View monthly trend chart

As Carlos **I want** a 12-month bar chart **So that** I can spot seasonal peaks

- **Priority:** Should · **Story points:** 2

Business rules

- Jan–Dec X axis; tCO_2e Y axis. Months with no data show 0 bars. Dashed average reference line. Current month highlighted.

Acceptance criteria

1. **Given** viewing 2026 in April **When** chart renders **Then** Jan–Apr show data and May–Dec are 0 bars.

US-024 · Switch the dashboard year

As Emma **I want** to switch between 2021–2026 **So that** I can review historical performance

- **Priority:** Must · **Story points:** 2

Business rules

- MUI `ToggleButtonGroup` year picker. 2021 labelled "Baseline" (green dot), 2026 labelled "Current" (amber dot).
- Changing year updates `?year=XXXX` URL and re-renders server component.

Acceptance criteria

1. **Given** on `/dashboard?year=2024` **When** I click "2025" **Then** URL changes and all panels refresh.

US-025 · View top emitting activities

As Emma **I want** a table of the top 5 emitting activities **So that** I know where to focus reductions

- **Priority:** Should · **Story points:** 2

Business rules

- Ranked by summed tCO_2e descending. Columns: rank badge (#1–#5), activity name, scope chip, tCO_2e, % of total.

US-026 · View consumption breakdown

As Carlos **I want** to see total consumption per unit **So that** I understand activity volumes

- **Priority:** Could · **Story points:** 2

Business rules

- Records grouped by `unit`. Each group: total quantity + top 3 contributing activities. Colour-coded unit chips.

9.5 Epic 4 — Emissions Calculator (Scopes 1–3)

Goal: Let users log activity data and have the system compute CO_2e automatically.

US-030 · Log a Scope 1 emission record

As Carlos **I want** to log fuel consumption **So that** our direct emissions are tracked

- **Priority:** Must · **Story points:** 5 · **Dependencies:** US-002

Validations

FIELD	RULE
Activity	Required · must belong to Scope 1 · `isActive = true`
Emission factor	Required · must belong to the selected activity
Quantity	Required · numeric · > 0 · max 15 significant digits
Unit	Required · matches activity default unit (informational)
Period	Required · month/year picker · must be ≤ current month
Notes	Optional · max 500 chars

Business rules

- CO_2e computed server-side: `co2eKg = quantity × factor.value`; `co2eT = co2eKg / 1000`.
- Rounded: `co2eKg` to 3 decimals, `co2eT` to 6 decimals.
- `EmissionRecord` stores both values (denormalised for fast reads).
- Selected `emissionFactorId` is immutable once saved — admin changes don't affect history.
- Successful save returns 201 with the full record.

Acceptance criteria

1. **Given** 12,400 kWh natural gas × DEFRA 2024 factor (0.18316 $kgCO_2e/kWh$) **When** I submit **Then** `co2eKg = 2,271.18` and `co2eT = 2.27118`.
2. **Given** quantity `-5` **When** I submit **Then** "Quantity must be greater than 0" — no record created.
3. **Given** future period (e.g. 2027-06) **When** I submit **Then** API returns 400 "Period cannot be in the future".
4. **Given** deactivated activity **When** form renders **Then** it doesn't appear in dropdown.

Edge cases

- Quantity 0 rejected.
- `Infinity` or `NaN` from DevTools rejected by Zod.

US-031 · Log a Scope 2 emission record

As Carlos **I want** to log monthly grid electricity consumption **So that** purchased-energy emissions reflect the correct grid intensity

- **Priority:** Must · **Story points:** 3

Business rules

- Same engine/validation as Scope 1.
- Factor picker defaults to `isDefault` factor but allows alternatives (UK, US, EU).
- Location-based method only in Phase 1.

Acceptance criteria

1. **Given** 48,000 kWh × 0.20493 $kgCO_2e/kWh$ (UK DEFRA 2024) **When** I submit **Then** `co2eKg = 9,836.64`.

US-032 · Log a Scope 3 emission record

As Carlos **I want** to log business travel, commuting, purchased goods, and waste **So that** our full value-chain footprint is captured

- **Priority:** Must · **Story points:** 3

Acceptance criteria

1. **Given** 24,600 passenger-km long-haul flight × 0.195 kgCO_2e/pkm **When** I submit **Then** `co2eKg = 4,797` and record appears in All Records.

US-033 · See factor details before saving

As Carlos **I want** to see the selected factor's source, region, and value before committing **So that** I don't use the wrong one

- **Priority:** Should · **Story points:** 1

Acceptance criteria

1. **Given** I select a factor **When** the side panel updates **Then** I see "Source: DEFRA · Region: UK · Value: 0.20493 kgCO_2e/kWh · GWP: IPCC AR5 100-year".

US-034 · See a live CO_2e preview

As Carlos **I want** the form to show estimated CO_2e before I submit **So that** I can sanity-check the magnitude

- **Priority:** Could · **Story points:** 2

Acceptance criteria

1. **Given** factor 0.20493 and quantity 1000 **When** I type **Then** "Preview" chip shows "≈ 204.93 kg CO_2e / 0.205 t CO_2e".

9.6 Epic 5 — Emission Records Management

Goal: Let tenant users review, filter, and manage what's been logged.

US-040 · View all records

As Emma **I want** a paginated, filterable list of every emission record **So that** I can audit my team's entries

- **Priority:** Must · **Story points:** 3

Business rules

- 20 records per page. Default sort: `period` descending.
- Columns: Date (period), Scope chip, Activity, Quantity + unit, tCO_2e, Emission factor, User, Data source.
- Scope chips colour-coded consistently.

Acceptance criteria

1. **Given** 450 records **When** I visit `/emissions` **Then** I see page 1 of 23, newest first.
2. **Given** I click Scope 2 filter **When** URL updates to `?scope=2` **Then** only Scope 2 records show.

US-041 · Filter records by year

As Emma **I want** to narrow records to a specific year **So that** I can produce period-specific audits

- **Priority:** Should · **Story points:** 1

Acceptance criteria

1. **Given** year dropdown on `/emissions` **When** I pick "2025" **Then** URL changes to `?year=2025` and list refreshes.

9.7 Epic 6 — Reporting

Goal: Turn raw records into ISO 14064-aligned annual reports.

US-050 · Generate an annual report

As Emma **I want** to generate a GHG report for a specific year **So that** I can share data with auditors, investors, or regulators

- **Priority:** Must · **Story points:** 8 · **Dependencies:** US-030, US-031, US-032

Validations

FIELD	RULE
Year	Required · 2021 ≤ year ≤ current year · must have ≥ 1 record · no existing report for that year

Business rules

- `POST /api/reports` with `{ year }`.
- API aggregates 4 parallel queries: all records for year, prior-year aggregate, baseline aggregate, entity metadata.
- Aggregated summary stored in `Report.summary` (JSONB).
- `Report.status` set to `GENERATED`.
- Duplicate (entity, year) → 409 Conflict.
- Zero records for year → 422 Unprocessable Entity.

Summary JSON shape:

```
{
  "totalCo2eT": 1248.5,
  "recordCount": 450,
  "scope1": { "co2eT": 342.1, "pct": 27.4, "activities": [] },
  "scope2": { "co2eT": 518.4, "pct": 41.5, "activities": [] },
  "scope3": { "co2eT": 388.0, "pct": 31.1, "activities": [] },
  "byMonth": [{ "month": 1, "co2eT": 92.3 }],
  "topActivities": [{ "name": "Grid electricity", "co2eT": 234.1, "scope"
  "baselineCo2eT": 1812.4,
  "prevYearCo2eT": 1418.0,
  "yoyChangePct": -11.95,
  "vsBaselinePct": -31.12
}
```

Acceptance criteria

1. **Given** I click "Generate Report" for 2025 **When** API completes **Then** `Report` row created with `status = GENERATED` and detail panel shown inline.
2. **Given** report for 2025 already exists **When** I try again **Then** "A report for 2025 already exists — delete it first".
3. **Given** 2027 has zero records **When** I try to generate **Then** "No emission records found for 2027".

US-051 · View report detail panel

As Emma **I want** to view the full breakdown of a report inline **So that** I can review without downloading

- **Priority:** Must · **Story points:** 5

Business rules

- Clicking a report row expands: 4 KPI cards (total, YoY, vs-baseline, data quality), scope bars, 12-month chart, top 5 sources.

- Data quality: High (≥ 50 records), Medium (20–49), Low (< 20).

US-052 · Export report as CSV

As Carlos **I want** to download a report as CSV **So that** I can analyse it in Excel

- **Priority:** Must · **Story points:** 3

Business rules

- `GET /api/reports/[id]/export?format=csv` returns `text/csv` with `Content-Disposition: attachment`.
- Filename: `carbonly-ghg-report-{year}.csv`.
- CSV blocks (in order): report header, summary, scope breakdown, activities by scope, monthly trend, top activities, footer credit.
- RFC 4180 escaping for commas, quotes, newlines.

US-053 · Publish / unpublish a report

As Emma **I want** to mark a report as PUBLISHED **So that** I signal internally when the report is ready to share

- **Priority:** Should · **Story points:** 2

Business rules

- `PATCH /api/reports/[id]` with `{ status: "PUBLISHED" | "GENERATED" }`. Publish only for GENERATED; unpublish only for PUBLISHED.

US-054 · Delete a report

As Emma **I want** to delete a mistakenly generated report **So that** I can re-generate with corrected data

- **Priority:** Should · **Story points:** 1

Business rules

- `DELETE /api/reports/[id]`. `confirm()` dialog warns action is irreversible.

9.8 Epic 7 — Settings & Profile

Goal: Let users review their profile and organisation details.

US-060 · View my profile

As Carlos **I want** to see my name, email, role, and member-since date **So that** I can verify the account I'm using

- **Priority:** Should · **Story points:** 1

US-061 · View my organisation details

As Emma **I want** to see my organisation's name, slug, industry, country, and subscription status **So that** I can confirm the tenant context

- **Priority:** Should · **Story points:** 1

US-062 · View billing summary

As Emma **I want** a Billing section showing my current plan, status, and trial end date **So that** I know where my subscription stands

- **Priority:** Must · **Story points:** 3

Business rules

- Plan resolved from `Entity.stripePriceId` via `getPlanByPriceId()`.
- Status chip colour: ACTIVE=green, TRIALING=blue, PAST_DUE=amber, CANCELED=red.
- If `stripeCustomerId` is set, "Manage billing" button opens Stripe Customer Portal.
- If NONE / CANCELED / TRIALING, "Upgrade plan" link routes to `/#pricing`.

Acceptance criteria

1. **Given** TRIALING subscription ending 2026-04-30 **When** I visit Settings **Then** I see "Trial ends in 15 days — 30 April 2026".

9.9 Epic 8 — Billing & Subscriptions

Goal: Monetise via Stripe with three tiers and a 14-day trial.

US-070 · View pricing on the landing page

As Priya **I want** to see three plan tiers with features and prices **So that** I can evaluate Carbonly

- **Priority:** Must · **Story points:** 3

Business rules

- Three cards: Starter £49/mo, Professional £149/mo (Most Popular badge), Enterprise (custom — contact sales).
- Annual billing option toggles monthly ↔ discounted annual prices.

US-071 · Start a paid plan via Stripe Checkout

As Emma **I want** to click "Start free trial" and go through Stripe Checkout **So that** my organisation starts a 14-day trial

- **Priority:** Must · **Story points:** 5

Validations

- User must be authenticated; if not, redirect to `/register`.
- `entityId` must exist.

Business rules

- `POST /api/stripe/checkout` with `{ planKey }`:
  1. Ensure Stripe Customer exists; save `stripeCustomerId`.
  2. Create Checkout Session: `mode: "subscription"`, `trial_period_days: 14`, `success_url: /dashboard?upgraded=1`, `cancel_url: /pricing?canceled=1`.
  3. Return `{ url }` for client redirect.
- On success, webhook → `subscriptionStatus = TRIALING`.

Acceptance criteria

1. **Given** I click "Start free trial" on Professional **When** API responds **Then** I'm redirected to Stripe Checkout with correct price ID and trial config.
2. **Given** I cancel at Stripe **When** I return **Then** I land on `/pricing?canceled=1` and no subscription was created.

US-072 · Keep subscription state in sync via webhooks

As the system **I want** to mirror Stripe state into the `Entity` table **So that** the UI reflects reality without polling

- **Priority:** Must · **Story points:** 5

Validations

- Webhook signature verified with `STRIPE_WEBHOOK_SECRET` via `stripe.webhooks.constructEvent()`.
- Unverified payloads → 400 with no DB writes.

Business rules — event mapping:

STRIPE EVENT	ACTION
`customer.subscription.created`	Save `subscriptionStatus`, `stripeSubscriptionId`, `stripePriceId`
`customer.subscription.updated`	Same as created
`customer.subscription.deleted`	Set `subscriptionStatus = CANCELED`
`invoice.payment_failed`	Set `subscriptionStatus = PAST_DUE`

Stripe status → Carbonly enum:

STRIPE	CARBONLY
trialing	TRIALING
active	ACTIVE
past_due	PAST_DUE
canceled	CANCELED
unpaid	UNPAID
incomplete / incomplete_expired / paused	PAST_DUE (fallback: CANCELED)

US-073 · Open the Stripe Customer Portal

As Emma **I want** a "Manage billing" button that opens Stripe's hosted portal **So that** I can update card, view invoices, and change plan

- **Priority:** Must · **Story points:** 2

Business rules

- `POST /api/stripe/portal` creates billing portal session with `return_url: /settings`. Returns `{ url }`.

9.10 Epic 9 — Admin: Entity Management

Goal: Let platform admins manage tenant organisations.

US-080 · View all entities

As Adam **I want** a searchable list of every tenant **So that** I can monitor and support them

- **Priority:** Must · **Story points:** 3

Business rules

- Table columns: Name, Slug (monospace), Industry, Country, Status chip, Users count, Records count, Created.
- Stat cards: Total, Active, Trialing, Churned.

US-081 · Create a new entity

As Adam **I want** to create a new organisation via a sidesheet form **So that** a new customer is onboarded

- **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Name	Required · 2–80 chars
Slug	Optional · auto-generated from name if empty · must be unique
Industry	Optional · free text
Country	Optional · free text
Subscription status	Defaults to `TRIALING`

Acceptance criteria

1. **Given** I type "GreenTech Solutions" **When** I submit without a slug **Then** entity created with slug `"greentech-solutions"`.

2. **Given** slug "acme-corp" already exists **When** I submit **Then** "Slug already in use".

US-082 · Edit an entity

As Adam **I want** to edit name, industry, country, or status **So that** I can keep records current

- **Priority:** Must · **Story points:** 2

US-083 · Delete an entity

As Adam **I want** to delete an entity and all its data **So that** I can offboard customers cleanly

- **Priority:** Should · **Story points:** 2

Business rules

- Prisma cascade deletes: `UserEntity`, `EmissionRecord`, `Report`, `YearlyConfig`.
- `confirm()` dialog warns "This will remove all users, records and reports".

Acceptance criteria

1. **Given** entity with 5 users, 450 records, 2 reports **When** I confirm deletion **Then** all rows removed and stat cards update.

9.11 Epic 10 — Admin: User Management

US-090 · View all users

As Adam **I want** a list of every registered user **So that** I can audit access

- **Priority:** Must · **Story points:** 2

Business rules

- Columns: Name, Email, Role chip, Organisation, Joined. "You" badge on current user.
- Stat cards: Total, Super Admins, Admins, Experts.

US-091 · Invite (create) a user

As Adam **I want** to create a new user and assign them to an entity **So that** they can sign in immediately

- **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Name	Optional · ≤ 80 chars
Email	Required · valid format · unique
Temporary password	Optional · if set, ≥ 8 chars
Global role	Required · EXPERT (default), ADMIN, or SUPER_ADMIN
Entity	Optional · if set, creates `UserEntity` row
Entity role	Required if entity is set · EXPERT (default) or ADMIN

Acceptance criteria

1. **Given** I invite `jane@acme-corp.com` as EXPERT at Acme Corp **When** I submit **Then** user + entity membership created.

US-092 · Edit a user

As Adam **I want** to edit a user's name or global role **So that** I can correct mistakes or promote users

- **Priority:** Must · **Story points:** 2

US-093 · Delete a user

As Adam **I want** to delete a user **So that** I can revoke access

- **Priority:** Must · **Story points:** 1

Business rules

- A user cannot delete their own account — API returns 400.
- Cascade deletes: `UserEntity`, `Session`, `Account`.

9.12 Epic 11 — Admin: Taxonomy Management

US-100 · View scopes & activities

As Adam **I want** to see each scope with its activities **So that** I can maintain the GHG Protocol taxonomy

- **Priority:** Must · **Story points:** 2

Business rules

- Three scope sections with coloured headers.
- Row: Activity name, description (truncated), Unit, Equation, Factor count, Active toggle.

US-101 · Add a new activity

As Adam **I want** to add a new activity under a scope **So that** I can keep up with reporting requirements

- **Priority:** Must · **Story points:** 2

Validations

FIELD	RULE
Scope	Required
Name	Required · ≤ 120 chars
Description	Optional · ≤ 500 chars
Unit	Required · ≤ 30 chars
Equation	Optional · defaults to "quantity × emissionFactor"

Business rules

- Auto-increments `sortOrder` = max+1 within scope. Defaults to `isActive = true`.

US-102 · Edit / deactivate an activity

As Adam **I want** to update an activity or toggle its active state **So that** I can retire deprecated activities

- **Priority:** Must · **Story points:** 2

Business rules

- Deactivated activities hidden from calculator dropdowns; existing records preserved.

US-103 · Delete an activity

As Adam **I want** to delete an activity with no factors **So that** I can clean up unused entries

- **Priority:** Should · **Story points:** 1

Business rules

- `DELETE` blocked with 409 if any `EmissionFactor` rows reference the activity.

9.13 Epic 12 — Admin: Emission Factor Management

US-110 · View all emission factors

As Adam **I want** a master table of every factor **So that** I can audit the taxonomy

- **Priority:** Must · **Story points:** 3

Business rules

- Columns: Scope chip, Activity, Factor name, Value (mono), Unit, Source, Region, Default chip.
- Search: filter by factor name, activity name, or source.
- Scope filter dropdown + stats (Total, Defaults, per-scope counts).

US-111 · Add a new factor

As Adam **I want** to add a new factor to an activity **So that** users have the latest regulator-issued values

- **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Activity	Required · must be active
Factor name	Required · ≤ 120 chars
Value	Required · positive number · ≤ 15 significant digits
Unit	Required (e.g. "kgCO2e/kWh")
Source	Optional
Region	Optional (e.g. "UK", "US", "EU")
GWP	Optional (e.g. "IPCC AR5 100-year")
Is default	Boolean

Business rules

- If `isDefault = true`, API automatically un-defaults all other factors for the same activity.

Acceptance criteria

1. **Given** "Grid electricity" has default factor "DEFRA 2023" **When** I add "DEFRA 2024" as default **Then** DEFRA 2023 is automatically un-defaulted in the same transaction.

US-112 · Edit / toggle-default a factor

As Adam **I want** to update a factor's value or toggle its default flag **So that** I can roll over to a new year's factor

- **Priority:** Must · **Story points:** 2

US-113 · Delete a factor

As Adam **I want** to delete a factor not referenced by any records **So that** I can clean up unused entries

- **Priority:** Should · **Story points:** 1

Business rules

- Returns 409 Conflict if any `EmissionRecord` rows reference the factor.

9.14 Epic 13 — Admin: Platform Analytics & Billing Insights

US-120 · View platform overview

As Adam **I want** a dashboard of platform-wide KPIs **So that** I can monitor health at a glance

- **Priority:** Must · **Story points:** 5

Business rules

- Hero banner (dark gradient) with welcome message.
- KPI cards: Organisations, Users, Emission Records, CO_2e Tracked (tonnes).
- Subscription status panel (5 tiles), taxonomy summary, recent orgs, recent users, activity feed.

US-121 · View billing analytics

As Adam **I want** revenue-focused analytics (MRR, conversions, trials ending) **So that** I can track SaaS metrics

- **Priority:** Must · **Story points:** 5

Business rules

- MRR = sum of `monthlyPrice` across entities with `subscriptionStatus = ACTIVE`.
- Conversion = % of entities currently `ACTIVE`.
- Trials ending = entities `TRIALING` with `trialEndsAt ≤ now + 7 days`.
- Plan distribution bars (Starter / Professional / Enterprise / None).
- Alert banner for trials ending soon.
- Full subscription table: plan, status, trial end, Stripe customer ID (truncated), user count.

Acceptance criteria

1. **Given** 3 active professional customers at £149/mo **When** I view `/admin/billing` **Then** MRR reads "£447".

9.15 Epic 14 — Marketing & Landing Page

US-130 · View the landing page

As Priya **I want** a polished landing page that explains Carbonly in 30 seconds **So that** I decide whether to sign up

- **Priority:** Must · **Story points:** 5

Business rules

- Sections: Hero (dark green gradient, typewriter headline) → Trusted-by chips → Animated stats bar → Features grid (6 items) → "How it works" (3-step auto-advance) → Pricing → Testimonials → CTA banner → Footer.
- Sticky nav transitions to frosted glass on scroll.
- Responsive: single column mobile, 2-col tablet, full layout desktop.
- All `/register` and `/login` links functional.

US-131 · Start free trial from pricing

As Priya **I want** "Start free trial" buttons on each plan card **So that** I can sign up directly from pricing

- **Priority:** Must · **Story points:** 2

Business rules

- Clicking calls `POST /api/stripe/checkout`; if unauthenticated (401), redirect to `/register`.
- Enterprise CTA opens `mailto:sales@carbonly.io?subject=Enterprise enquiry`.

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10. Cross-cutting Acceptance Criteria

The following acceptance criteria apply to **every** story in this document:

10.1 Security

#	CRITERION
SEC-01	No tenant data is ever returned or mutated without filtering by `entityId` from the session
SEC-02	All destructive admin actions require `confirm()` on the client
SEC-03	Stripe webhooks are signature-verified before any DB mutation
SEC-04	No secret (Stripe key, NextAuth secret, DB password) is exposed in the client bundle
SEC-05	Server-sent HTML escapes all user-supplied strings (React defaults)
SEC-06	Middleware enforces auth at the edge — no route can be accessed with an invalid/missing session

10.2 Error Handling

#	CRITERION
ERR-01	All API routes return `{ error: string }` JSON with appropriate 4xx/5xx status
ERR-02	Client UI displays errors using MUI `Alert` components or inline helper text
ERR-03	Network failures show "Network error — please try again"
ERR-04	500-level errors are logged server-side (Prisma error at `error` level in production)

10.3 Accessibility

#	CRITERION
A11Y-01	All interactive elements are keyboard accessible (Tab order, Enter/Space to activate)
A11Y-02	Form inputs have associated `<label>` elements
A11Y-03	Colour is never the sole indicator of state (badges include text + icon)
A11Y-04	Focus rings are visible on all focusable elements

10.4 Performance

#	CRITERION
PERF-01	Initial page load ≤ 1.5s on fast 3G
PERF-02	Dashboard data query ≤ 50ms on ~2,000 records
PERF-03	All pages use server components; no client-side data fetch on first load

10.5 Responsive Design

#	CRITERION
RESP-01	All pages render correctly at 375px (mobile), 768px (tablet), 1280px (desktop)
RESP-02	Admin tables support horizontal scroll on narrow viewports (`table-responsive`)
RESP-03	Sidebar is permanent drawer on ≥ 992px; temporary hamburger drawer below

10.6 Data Integrity

#	CRITERION
DATA-01	All FK relationships use cascading delete where ownership is clear
DATA-02	Emission factor references in `EmissionRecord` are immutable post-creation
DATA-03	All CO_2e calculations are performed server-side; client inputs are never trusted
DATA-04	`EmissionRecord` captures `userId`, `dataSource`, and `createdAt` for full auditability

---

11. Requirements Traceability Matrix

Each functional requirement is covered by one or more user stories:

BRD REF	REQUIREMENT	USER STORIES
FR-AUTH-01	Email/password + Google auth	US-001, US-002, US-003
FR-AUTH-02	Session expiry	US-005
FR-AUTH-03	SUPER_ADMIN platform access	US-080 – US-121
FR-AUTH-04	Entity ADMIN manages their entity	US-061, US-062, US-073
FR-AUTH-05	EXPERT scoped to assigned entity	US-030, US-031, US-032
FR-AUTH-06	Onboarding redirect	US-010
FR-AUTH-07	Middleware edge auth	US-005
FR-AUTH-08	bcryptjs password hashing	US-001 (business rules)
FR-CALC-01	Scope 1/2/3 calculators	US-030, US-031, US-032
FR-CALC-02	Activity + factor selection	US-033
FR-CALC-03	`CO2e = quantity × factor`	US-030 (business rules)
FR-CALC-04	Store kg + tonnes	US-030 (business rules)
FR-CALC-05	Capture period + notes	US-030 (validations)
FR-CALC-06	Factor source/region/GWP	US-033, US-110
FR-CALC-07	Immutable factor reference	US-030 (business rules)
FR-CALC-08	No future periods	US-030 (AC-3)
FR-CALC-09	Server-side calculation	§10.6 DATA-03
FR-DASH-01	Total by scope hero	US-020, US-021
FR-DASH-02	Donut/pie chart	US-022
FR-DASH-03	Monthly trend chart	US-023
FR-DASH-04	Top 5 activities	US-025
FR-DASH-05	Year selection	US-024
FR-DASH-06	YoY + vs-baseline	US-020
FR-DASH-07	Scope drill-down	US-021
FR-DASH-08	Consumption breakdown	US-026
FR-DASH-09	RSC-first rendering	§10.4 PERF-03

BRD REF	REQUIREMENT	USER STORIES
FR-ENT-01	Entity CRUD	US-080, US-081, US-082, US-083
FR-ENT-02	Unique slug	US-081 (validations)
FR-ENT-03	Edit entity profile	US-082
FR-ENT-04	Stripe subscription mirror	US-072
FR-ENT-05	Entity cascade delete	US-083
FR-USR-01	View all users	US-090
FR-USR-02	Invite/create user	US-091
FR-USR-03	Change role	US-092
FR-USR-04	Delete user (no self-delete)	US-093
FR-SCOPE-01	Activity CRUD	US-100, US-101, US-102, US-103
FR-SCOPE-02	Activity metadata	US-101 (validations)
FR-SCOPE-03	Block delete if factors exist	US-103
FR-SCOPE-04	Deactivation preserves history	US-102
FR-EF-01	Factor CRUD	US-110, US-111, US-112, US-113
FR-EF-02	Single default per activity	US-111 (business rules)
FR-EF-03	Auto-undefault on new default	US-111 (AC-1)
FR-EF-04	Block delete if records exist	US-113
FR-PAY-01	Three subscription plans	US-070
FR-PAY-02	Stripe Checkout + 14-day trial	US-071
FR-PAY-03	Webhook status sync	US-072
FR-PAY-04	Lapsed → billing redirect	US-062
FR-PAY-05	Stripe Customer Portal	US-073
FR-PAY-06	Webhook signature verification	US-072 (validations)
FR-RPT-01	Generate annual report	US-050
FR-RPT-02	Scope + activity breakdown	US-051

BRD REF	REQUIREMENT	USER STORIES
FR-RPT-03	CSV export	US-052
FR-RPT-04	Status lifecycle	US-053
FR-RPT-05	Block duplicate report	US-050 (AC-2)
FR-RPT-06	422 on zero records	US-050 (AC-3)
NFR-01	Tenant isolation	§10.1 SEC-01
NFR-02	RSC-first rendering	§10.4 PERF-03
NFR-03	Performance thresholds	§10.4 PERF-01, PERF-02
NFR-04	99.9% availability	§7 (Enterprise SLA)
NFR-05	500+ entity scalability	Shared schema architecture
NFR-06	Audit fields	US-030, §10.6 DATA-04
NFR-07	Factor traceability	US-110, US-111
NFR-08	Accessibility	§10.3
NFR-09	Responsive design	§10.5
NFR-10	Immutable factor refs	§10.6 DATA-02
NFR-11	Structured error responses	§10.2 ERR-01

12. Glossary

TERM	DEFINITION
GHG	Greenhouse Gas
CO_2e	Carbon Dioxide equivalent — standard unit for measuring carbon footprints
tCO_2e	Metric tonnes of CO_2 equivalent
Scope 1	Direct emissions from owned or controlled sources (combustion, fugitive)
Scope 2	Indirect emissions from purchased energy (electricity, heat, steam)
Scope 3	All other indirect emissions in the value chain (travel, goods, waste)
EF	Emission Factor — the CO_2e emitted per unit of activity (e.g. $kgCO_2e/kWh$)
GWP	Global Warming Potential — multiplier for converting non-CO_2 gases to CO_2e
Entity	A tenant organisation on the Carbonly platform
RSC	React Server Component — a Next.js 13+ component that renders server-side only
JWT	JSON Web Token — the session format used by NextAuth v5
MRR	Monthly Recurring Revenue
SUPER_ADMIN	Platform-level administrator with access to all entities
ADMIN	Entity-level administrator managing one or more organisations
EXPERT	Carbon expert end user who enters and reviews emission data
YoY	Year-over-year change expressed as a percentage
GHG Protocol	The Greenhouse Gas Protocol Corporate Accounting and Reporting Standard
DEFRA	UK Department for Environment, Food & Rural Affairs — source of UK emission factors
EPA	US Environmental Protection Agency — source of US emission factors
IPCC AR5	IPCC Fifth Assessment Report — GWP values (100-year time horizon)
TCFD	Task Force on Climate-related Financial Disclosures

TERM	DEFINITION
CDP	Carbon Disclosure Project
EU CSRD	EU Corporate Sustainability Reporting Directive
SaaS	Software as a Service
MoSCoW	Prioritisation framework: Must / Should / Could / Won't

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